# Robot Autonomy - Homework 3

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This homework will focus on task planning and grasping. We will be defining and solving problems pertaining to the arrangement of blocks using symbolic planning framework we learnt in class. We will also analyze grasp feasibility from a mathematical modeling point of view.

## 1 -Symbolic Planning

This problem set introduces two files: **SymScript.py** and **SymPlanner.py**. SymPlanner includes code for a domain-independent symbolic planner. SymScript defines specific domains and queries. We will be updating both files using concepts we learnt in class.

### 1.1 - Block World Domain

Start by examining the make_Room_domain() function in SymScript.py. It specifies actions using the ActionSchema() class in SymPlanner.py that can be used with the Planner() class functions.

Similarly, define actions in the make_Block_domain() function that can be used to solve problems in the Block World. Think about the actions that need to be performed to move blocks around like you did in Lab1.

NOTE: You can't solve the problem yet because we haven't defined a way to plan about the states and actions!

### 1.2 - Discrete Search

Next, you will implement A* search to find a plan of actions to solve the problem using the Planner() class functions. The problem has been setup in the plan() function. You must fill in the remaining code.

Once you have successfully implemented the search algorithm, you will be able to solve the Room problem by running SymScript.py.

### 1.3 - Planning In Block World

Now that you have defined the action space for the Block World and you have written a working discrete search planner about states and actions, you can now solve problems in the Block World!

Write a block of code in the main function of SymScript.py using the Room problem block as an example to solve the Block problem. You will use the action domain you have defined in part 1.1 and the planner in part 1.2.

### 1.4 - Another Block World Problem

We are now going to create a new problem for the Block World domain. The initial and goal states are shown in Fig. 1.

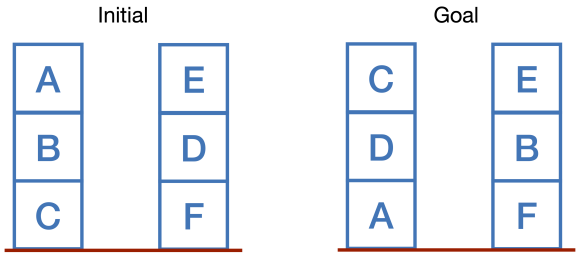



Figure 1: Another Block World Problem

Add the initial and goal states to make_Block2_problem() in SymScript.py. Solve the problem in the same way you solved the first two problems.

### 1.5 - Submission

Submit your SymPlanner.py and SymScript.py files along with screenshots of the plans generated for the two Block World problems in your writeup.

## 1 2 - Grasping

In this section we are going to explore form closure and grasp metrics.

### 2.1 - Removing Contacts

Fig. 2 shows a 2D $(X, Y, \theta)$ example of a form-closure grasp with six frictionless contacts. Write out the grasp matrix G for this grasp. Plot the grasp vectors g using an F-$X$ vs. T-$\theta$ plot and an F-$Y$ vs. T-$\theta$ plot.

In our lecture example, we found a solution using only four contacts to achieve a form-closure grasp of a box. Without shifting any of the contact points, can any of the contact points be removed while still maintaining form closure? Explain your answer in 1–2 sentences.

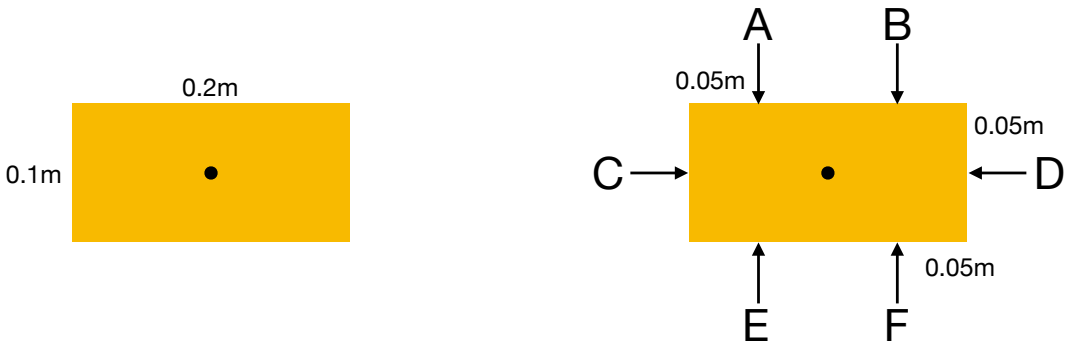



Figure 2: Object and grasp for problem 2.1

### 2.2 - Grasp Metrics

Consider the triangle shape in Fig. 3. You need to choose one of the three possible grasps (A, B, or C) to execute. Each grasp consists of two contacts on the slanted edges (e.g., B1 and B2 for the B grasp) with a coefficient of Coulomb friction of $\mu = 1$, as well as a shared contact at S that is *frictionless*. Grasp A contacts are at the top corner, grasp C contacts are at the bottom corners, and grasp B contacts are exactly in the middle of the sides. For each grasp, write out the grasp matrix G and compute the grasp quality metric $\sqrt{\det(GG^T)}$. Based on this grasp metric, which contacts provide the best grasp A, B, or C? Explain your answer in 1-2 sentences.

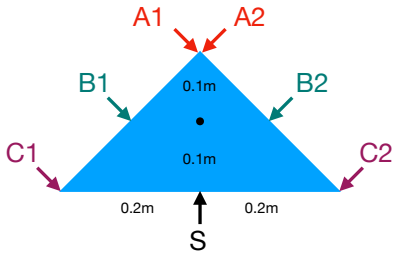



Figure 3: Object and grasps for problem 2.2

## 2 Submission

- The submitted zip file should unzip into a zip file named andrewid_hw3_code_16662.zip AND one pdf named andrewid_hw3_16662.pdf with your solutions.
- Please ensure your venv or conda environment is NOT submitted in the zip folder.
- Kindly upload your zip file respecting the following naming convention: andrewid_hw3_16662.zip